

Physics - Work, Energy & Power

1. (a) The work done by 2.40-N push

$$\Rightarrow W = Fd \cos(\theta)$$

$$\theta = 0$$

$$W = 2.4 \text{ N} \times \cos(0) \quad d = 1.5$$

$$W = 2.4 \times 1.5 \times \cos(0)$$

$$= 3.60 \times 1 = 3.60 \text{ J}$$

(b) Work done by friction force

$$W = Fd \cos \theta \quad \theta = 180^\circ$$

$$= 0.600 \times 1.50 \times \cos(180^\circ)$$

$$= 0.600 \times 1.50 \times -1$$

$$= -0.90 \text{ J}$$

(c) The Normal force from table top is perpendicular to the direction of motion so $\theta = 90^\circ$

$$\cos \theta = \cos(90^\circ) = 0$$

$\therefore$ Work done by normal force is zero

$$\text{Work Done}_N = 0.0 \text{ J}$$

d) Similarly, gravity also acts perpendicular to the direction of motion, so the work done by gravity is also zero

$$\text{Work Done}_g = 0.0 \text{ J}$$

c) The net work done on the book is the sum of the work done by all the forces. So

$$\begin{aligned}\text{Net work Done} &= 3.60 \text{ J} + (-0.90 \text{ J}) + 0 + 0 \\ &= 2.70 \text{ J}\end{aligned}$$

2. Given : Mass of painter $m = 75.0 \text{ kg}$
Length of ladder $L = 2.75 \text{ m}$
Angle with the wall $\theta = 30^\circ$

$$\begin{aligned}h &= L \cos \theta = 2.75 \text{ m} \cdot \cos 30^\circ = 2.75 \text{ m} \cdot \frac{\sqrt{3}}{2} \approx 2.378 \text{ m} \\ &\approx 2.378 \text{ m}\end{aligned}$$

Work Done by gravity

$$W = -mgh \quad g = 9.8 \text{ m/s}^2$$

$$\begin{aligned}W &= -75.0 \text{ kg} \times 9.8 \text{ m/s}^2 \times 2.378 \text{ m} \\ &= -1747.71 \text{ J}\end{aligned}$$

$$W \approx -1750 \text{ J}$$

The Work Done by gravity ~~is~~ on the painter
 $\Rightarrow -1750 \text{ J}$

The answer does not depend on whether the painter climbs at constant speed or accelerates, because W.D by gravity only depends on the vertical displacement.

Q. Given: Distance travelled by car $d = 48 \text{ m}$
 mass of the car $m = 380 \text{ kg}$
 constant force $\vec{F} = (-68.0 \text{ N})\hat{i} + (36.0 \text{ N})\hat{j}$

$$W = F \cdot s \cos \phi$$

$$= \vec{F} \cdot \vec{s}$$

$\phi = 0$ (if $\vec{F}$ & $\vec{s}$ are in same direction)

$\phi = 180$ (if $\vec{F}$ & $\vec{s}$ are in opposite direction)

$$\vec{s} = s_x \hat{i} + s_y \hat{j}$$

$$s_x = 48.0 \cos(240^\circ) = -24.0 \text{ m}$$

$$s_y = 48.0 \sin(240^\circ) = -41.6 \text{ m}$$

$$\vec{s} = (-24.0 \hat{i} - 41.6 \hat{j}) \text{ m}$$

$$W = \vec{F} \cdot \vec{s}$$

$$= (-68.0 \hat{i} + 36.0 \hat{j}) \text{ N} \cdot (-24.0 \hat{i} - 41.6 \hat{j}) \text{ m}$$

$$= (-68 \times -24) + (36 \times -41.6)$$

$$= 1632 - 1498 = 134 \text{ J in } +x\text{-axis}$$

4: By Work-Energy Theorem
~~EA~~ $K_i =$ Energy lost due to gravity
& Kinetic friction

$$S = \frac{h}{\sin \alpha}$$

$$W_g = mgh$$

$$f_k = \mu_k mg \cos \alpha$$

$$W_f = -f_k S = -\mu_k mg \cos \alpha \cdot \frac{h}{\sin \alpha}$$

$$= -\mu_k mgh \tan \alpha$$

$$K_{\text{top}} - K_{\text{bottom}} = W_g + W_f \quad K_{\text{top}} = 0$$

$$-\frac{1}{2}mv^2 = mgh - \mu_k mgh \tan \alpha$$

$$\frac{1}{2}v^2 = gh(1 - \mu_k \tan \alpha)$$

$$v^2 = 2gh(1 - \mu_k \tan \alpha)$$

$$v = \sqrt{2gh(1 - \mu_k \tan \alpha)}$$

$$v_{\text{min}} = \left(2gh(1 - \mu_k \tan \alpha)\right)^{\frac{1}{2}}$$

$$5. \quad W = \Delta K = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \quad (\text{Work-Energy Theorem})$$

$$W = \frac{1}{2} m (|\vec{v}_f|^2 - |\vec{v}_i|^2)$$

$$= \frac{1}{2} (30) (5.62^2 - 3.90^2) \quad \begin{array}{l} m = 30.0 \text{ kg} \\ v_i = 3.90 \text{ m/s} \\ v_f = 5.62 \text{ m/s} \end{array}$$

$$= 15 (31.61 - 15.2)$$

$$= 15 \times 16.4 = 246 \text{ J}$$

The change in direction does not affect the work, because K.E depends only on speed.

$$1. \quad P_{\text{req}} = m_{\text{total}} g v = (600 + 65n) \cdot 9.8 \times 1.25 \\ = 12.25 (600 + 65n) = 7350 + 796.25n$$

$$P_{\text{req}} \leq 29840$$

$$7350 + 796.25n \leq 29840$$

$$n \leq \frac{29840 - 7350}{796.25} \approx 28.24$$

$\therefore$ Maximum passengers = 28 passengers

6.

Work-Energy theorem

$$W = \Delta K = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$

box starts at rest $v_0 = 0$

$$\therefore W = \frac{1}{2}mv^2$$

$$F(x) = 18.0 - (0.530)x \quad \text{+ x direction}$$

$$W = \int_0^{14.0} F(x) \cdot dx$$

$$= \int_0^{14} (18.0 - (0.530)x) dx$$

$$= \left[18x - \frac{0.530x^2}{2} \right]_0^{14}$$

$$= 18(14) - 0.530 \frac{(14)^2}{2} = 0$$

$$= 252 - 0.265 \times 196$$

$$= 252 - 51.9 \approx 200.1 \text{ J}$$

$$\frac{1}{2}mv^2 = 200.1 \text{ J}$$

$$6.00 V^2 = 200.1 \times 2$$

$$V^2 = \frac{400.2}{6.00}$$

$$V = \sqrt{66.7} \approx 8.17 \text{ m/s}$$

8. a) $F(x) = -ax^2$; SI unit of $F = N$
 SI unit of $x^2 = m^2$
 $\therefore$ SI unit $a = \frac{N}{m^2}$

b) Work from $(0.10\text{ m}, 0)$ to $(0.30\text{ m}, 0)$

$$W_{0.10 \rightarrow 0.30} = \int_{0.10}^{0.30} F_x dx = \int_{0.10}^{0.30} (-ax^2) dx$$

$$= -a \left[\frac{x^3}{3} \right]_{0.10}^{0.30}$$

$$= -\frac{12}{3} \left((0.30)^3 - (0.10)^3 \right)$$

$$= -\frac{12}{3} (0.027 - 0.001)$$

$$= -4 (0.026) = -0.104\text{ J}$$

c) Work from $(0.30\text{ m}, 0)$ to $(0.10\text{ m}, 0)$

$$W_{0.30 \rightarrow 0.10} = \int_{0.30}^{0.10} F_x dx = - \int_{0.10}^{0.30} F_x dx$$

$$\left(\int_a^b f(x) = - \int_b^a f(x) \right)$$

$$= -(-0.104) = +0.104\text{ J}$$

d) Yes, It depends on position x & is derivable from a scalar potential $F_x = -\frac{dU}{dx}$

$$U(0) = 0$$

$$F_x = -\frac{dU}{dx} = -ax^2 \Rightarrow \frac{dU}{dx} = ax^2$$

$$U(x) = \int ax^2 dx = \frac{ax^3}{3} + C$$

$$U(0) = 0 \Rightarrow C = 0$$

$$\therefore U(x) = \frac{a}{3}x^3$$

9. a) $U_i = mgh$ $K_i = 0$

zero potential at the bottom $\Rightarrow U_f = 0$ $K_f = \frac{1}{2}mv^2$

$$mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh}$$

vertical drop $h = d \sin \alpha$

$$v = \sqrt{2g(d \sin \alpha)}$$

$$v = (2gd \sin \alpha)^{1/2}$$

b) zero potential at the top $U_i = 0$ $K_i = 0$
 $U_f = -mgh$ $K_f = \frac{1}{2}mv^2$

$$\frac{1}{2}mv^2 - mgh = 0$$

$$\frac{1}{2}mv^2 = mgh$$

$$v = \sqrt{2gh} = \sqrt{2gdsin\alpha}$$

c) There was no work because the normal force is always perpendicular to the motion along the ramp, so it does no work. Only gravity changes mechanical energy. The normal force merely constrains the motion without adding or removing energy.

$$v = \sqrt{2gdsin\alpha}$$

10. $m = 2.50 \text{ kg}$ $k = 25.0 \text{ N/cm} = 2500 \text{ N/m}$
 $U = 11.5 \text{ J}$

$$\frac{1}{2} m v_{\max}^2 = U \Rightarrow v_{\max} = \sqrt{\frac{2U}{m}}$$

(a) $v_{\max} = \sqrt{\frac{2(11.5)}{2.50}} = \sqrt{9.2} \approx 3.03 \text{ m/s}$

$$U = \frac{1}{2} k x_{\max}^2 \Rightarrow x_{\max} = \sqrt{\frac{2U}{k}}$$

$$x_{\max} = \sqrt{\frac{23.0}{2500}} \approx 0.096 \text{ m}$$

b) $a = \frac{F}{m} = \frac{kx}{m}$

$$a_{\max} = \frac{k x_{\max}}{m}$$

$$= \frac{2500 (0.096)}{2.50}$$

$$= 96.0 \text{ m/s}^2$$

(c) The greatest acceleration occurs at the release moment (maximum compression), directed outward (opposite the compression). Acceleration decreases linearly with x & becomes 0 at spring's natural length where speed is maximal.